

Sarah Alnegheimish

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Research Interests: Machine Learning Frameworks, Deep Learning, Time Series, Foundation Models.

EDUCATION

2022 - 2025	Massachusetts Institute of Technology Ph.D. in Electrical Engineering and Computer Science (EECS) <i>Supervised by Dr. Kalyan Veeramachaneni</i>	GPA: 5.0/5.0
2019 - 2022	Massachusetts Institute of Technology S.M. in Electrical Engineering and Computer Science (EECS) S.M. in Computational Science and Engineering (CSE) <i>Supervised by Dr. Kalyan Veeramachaneni</i>	GPA: 5.0/5.0
2013 - 2017	King Saud University B.Sc. in Computer Science <i>First Class Honors</i>	GPA: 5.0/5.0

EXPERIENCE

Research Assistant | DAI Lab, MIT *Cambridge, MA*
Foundation Models for Time Series *Oct 2023 - present*

- Proposed *SigLLM*, a framework for probing large language models to solve complex time series tasks, specifically time series anomaly detection, and tested its efficacy.
- Built a benchmarking framework to conduct comprehensive evaluations of pre-trained foundation models for unsupervised time series anomaly detection, with regular releases and leaderboards.

Machine Learning Frameworks for Time Series *Sep 2022 - present*

- Leading *Sintel* ecosystem for time series modeling and analysis.
- Designed and implemented universal representations of *primitives* and *pipelines* for time series tasks including: classification, forecasting, and anomaly detection.
- Conducted real-world case studies featuring two predictive maintenance studies on wind turbine data and two studies on anomaly detection for satellite data and electric vehicle data.

Unsupervised Time Series Anomaly Detection (Master Thesis) *Sep 2019 - May 2022*

- Developed *Orion*, an open-source library for unsupervised time series anomaly detection ([github stars: 1,000](#)).
- Collaborated with students on a number of anomaly detection pipelines including: an autoencoder with regression model (AER) and a generative adversarial model (TadGAN) for reconstruction-based anomaly detection.

Gender Bias in Large Language Models *Sep 2021 - Feb 2022*

- Created a new dataset to examine model components that elicit biased output.
- Proposed mitigation strategies to reduce the discrimination severity of the model output.

Intuitive Physics for Game Play Bots *Sep 2019 - May 2020*

- Built a probabilistic program to infer actions performed by an agent in Newtonian settings.
- Complemented the program with a model-free approach for efficient sampling.

Applied Scientist Intern | Amazon *Seattle, WA*
Multivariate Time Series Anomaly Detection *May 2024 - Aug 2024*

- Developed a multivariate time series model to detect anomalies from multiple systems pertaining to a single asset.
- Packaged the model in a domain-specific framework for seamless end-to-end usage and benchmarking.

Software Engineer Intern | DataCebo *Boston, MA*
Modeling Generative Time Series *Jun 2021 - Aug 2021*

- Worked on improving the modularization of the PARModel which is a generative model for synthetic time series.
- Worked closely with the SDV team on reversible data transforms and conditional sampling.

Research Specialist | Center for Complex Systems at KACST and MIT *Riyadh, Saudi Arabia*
Predictive Framework for Healthcare Data *Dec 2017 - Aug 2019*

Collaborated with Dr. Kalyan Veeramachaneni

- Developed *Cardea*, an automated machine learning library to solve health related prediction problems.
- Integrated HL7's *Fast Healthcare Interoperability Resources* standard as a representation for data.

Occupation-Skill Space

Dec 2017 - Aug 2019

Collaborated with Professor Iyad Rahwan

- Constructed an occupation network by discovering job-job relationships through their underlying skills.
- Utilized the job space as a function to predict individual career trajectories.
- Analyzed the impact of automation on job mobility and the education hierarchy.

Data Analyst | Mozn

Riyadh, Saudi Arabia

Fraud Detection and Optical Character Recognition

May 2017 - Dec 2017

- Worked closely with a team of experts to develop a system that detects fraudulent shipments.
- Curated a dataset of Arabic characters and trained a convolutional model for automated character recognition.

SELECTED PUBLICATIONS

- **Alnegheimish, S.**, Nguyen, L., Berti-Equille, L. and Veeramachaneni, K., *Can Large Language Models be Anomaly Detectors for Time Series?*. IEEE International Conference on Data Science and Advanced Analytics (DSAA), 2024.
- **Alnegheimish, S.**, Berti-Equille, L., and Veeramachaneni, K., *OrionBench: Benchmarking Time Series Generative Models in the Service of the End-User*. IEEE International Conference on Big Data, 2024.
- **Alnegheimish, S.**, Liu, D., Sala, C., Berti-Equille, L. and Veeramachaneni, K., *Sintel: A Machine Learning Framework to Extract Insights from Signals*. ACM SIGMOD International Conference on Management of Data, 2022.
- **Alnegheimish, S.***, Guo, A.* , Sun, Y.* , *Using Natural Sentence Prompts for Understanding Biases in Language Models*. Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2022.
- Alhasoun, F. and **Alnegheimish, S.**, *Probabilistic Programming Bots in Intuitive Physics Game Play*. 35th AAAI Conference on Artificial Intelligence, 2021.
- Geiger, A., Liu, D., **Alnegheimish, S.**, Cuesta-Infante, A., and Veeramachaneni, K., *TadGAN: Time Series Anomaly Detection Using Generative Adversarial Networks*. IEEE International Conference on Big Data, 2020. (citations: 345).

HONORS & AWARDS

- MIT's EECS Graduate Alumni Fellowship, 2022-2023.
- Graduate Scholarship, King Abdulaziz City for Science and Technology (KACST), 2019 - 2022.
- Google's CS Research Mentorship Program (CSRMP), 2020 - 2021.
- Graduate Fellow, MiSK, 2019 - 2022.
- 2nd place at AEC annual best graduation project, 2017.
- Best graduation project in the college of Computer and Information Sciences, King Saud University, 2017.
- Best poster at the 9th Undergraduate Research Conference, Zayed University, 2017.
- Dean's list for outstanding students at King Saud University, 2015 and 2016.

SKILLS

Programming Languages	Python, Tensorflow, Pytorch, MATLAB, Java, R, C/C++, JavaScript, HTML, SQL English (native), Arabic (native)
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MEDIA

VentureBeat

- LLMs can't outperform a technique from the 70s, but they're still worth using. October 13, 2024

MIT News

- MIT researchers use large language models to flag problems in complex systems. August 14, 2024
- One-stop machine learning platform turns health care data into insights. April 9, 2021
- Method finds hidden warning signals in measurements collected over time. December 17, 2020